

Course Code	Name of the Course				
216U03C401	Probability, Statistics and Optimization Techniques				
Teaching Scheme	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course pre-requisites:

- Basics of Linear Algebra and Calculus

Course Objectives:

The main objective of the course is to provide education to students so that they can meet the growing need for qualified personnel in the fields of data science. The course content is designed to develop theoretical and practical literacy in computer science and statistics, with relevance in data science, so that students are able to critically select and apply appropriate methods and techniques to extract relevant and important information from data.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Apply concepts of continuous and discrete distributions to solve Engineering problems.
- CO2.** Learn statistical concepts for data summarization and data visualization.
- CO3.** Understand sampling and parameter estimation for statistical analysis.
- CO4.** Conduct hypothesis tests including Z-test, t-test, and F-test and use them in Statistical Model development.
- CO5.** Learn and apply Regression and Optimization approaches for problem solving.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Probability		04	CO1
	1.1	Conditional and Joint Probability, Bayes' Theorem, PDF & CDF of Continuous and Discrete random variable, Binomial Distribution, Poisson Distribution, Geometric Distribution, Uniform Distribution, Exponential Distribution, Normal Distribution, Chi-Square Distribution, Student's t-Distribution, F-Distribution		
2	Statistical Data Analysis		08	CO2
	2.1	Data types and Data pre-processing techniques		
	2.2	Data summarization using Descriptive Statistics, Mean, Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Skewness, Kurtosis, Mode, Range, Variance, Standard Deviation, Percentile, IQR, MAD, Correlation, Covariance		
	2.3	Data visualization as a analytical tool using basic graphs, Histogram, Bar plot, Line plot, Scatter plot, Box plot, Stem and Leaf plot, Pie chart, Heat map, Density and Contour plots, Data visualization as a medium of communication		
3	Sampling and Estimation		12	CO3
	3.2	Random variable, Statistical Inference, Population Parameter and Sample Statistic. Sampling Distribution, Sampling distribution of mean and variance, Central Limit Theorem		
	3.3	Population Parameter Estimation: Introduction, Maximum Likelihood Estimation(MLE) method for point estimation of population parameter, MLE to estimate parameter of Binomial, Poisson, Gaussian and Uniform distribution		
	3.4	Confidence Interval method for population parameter estimation: Two sided and One sided confidence Interval estimates, For a Normal mean when variance of population is known, For a Normal mean when variance of population is not known, For variance of Normal distribution		
4	Hypothesis testing		06	CO4
	4.1	Introduction, Setting up a hypothesis test, Null Hypothesis and Alternative Hypothesis, One tailed and two tailed test, Type I and type II errors		
	4.2	Hypothesis testing for Population Mean with known variance: z-test, Hypothesis testing for Population Mean under unknown variance: t-test		
	4.3	ANOVA: Introduction, One-way ANOVA, Two-way ANOVA		

5				
Regression and Optimization				
	5.1	Regression: Supervised learning, Linear Regression, Simple Linear Regression (SLR), Estimation of SLR model parameters using Ordinary Least Squares (OLS) method; Multiple Linear Regression (MLR) model, Estimation of model parameters using OLS Method, Multi-collinearity; Interpretation of SLR and MLR model parameters, model assessment using model parameters, Validation of assumptions in model using diagnostic plots; Regression Model building for data consisting of quantitative and qualitative variables	15	CO5
	5.2	Optimization basics; Optimization for convex functions, Gradient Descent method; Optimization for non-differentiable functions, subgradient descent method; Newton's method for second order differentiable functions; Constrained optimization using Lagrangian method		
Total			45	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Sheldon M. Ross	<i>Probability and Statistics for Engineers and Scientists</i>	Academic Press	4 th edition, 2014
2	Jiawei Han, Micheline Kamber, Jian Pei	<i>Data Mining: Concepts and Techniques</i>	Morgan Kaufmann	3rd edition 25 July 2011
3	Jake VanderPlas	<i>Python Data Science Handbook: Essential Tools for Working with Data</i>	O'Reilly	1 st edition, January 2016
4	Howard J. Seltman	<i>Experimental Design and Analysis</i>		July 2018
5	Douglas C. Montgomery, Elizabeth A. Peck, G. Geoffrey Vining	<i>Introduction to Linear Regression Analysis</i>	WILEY	6 th edition, February 2021
6	J. K. Sharma	<i>Operation research: Theory and Applications</i>	Laxmi Publications, India	6th Edition 2017